

factors require *dynamic* trading involving *long-short* positions, where we constantly have to adjust portfolio weights. Just like their static factor cousins, dynamic factor strategies—like value-growth investing, momentum, and short volatility strategies—don’t always make money, and investors have to brace for stomach-turning stumbles. It is precisely for suffering these losses that factors accrue risk premiums. Assets embed different combinations of factor risks, just as foods mix different types of nutrients.

The simplest factors are the equity and bond indexes that Norway selects as its benchmark returns. It’s easy to invest in these factors by buying cheap index funds.⁴ The traditional approach to asset management has been to build on an equity-bond mix by adding other asset classes, notably subclasses of equities and bonds such as emerging markets and high-yield bonds, or *alternative assets* like real estate, private equity, and hedge funds. (This is referred to, incorrectly in the context of its original incarnation by Swensen (2009), as the *endowment approach*; see chapter 13.⁵) Even without adding alternative asset classes, the equity-bond factor decision is the most important one taken by funds. And it explains the majority of the variation in performance.

2.2. RETURN DECOMPOSITION

To be concrete, I denote r as a fund’s return. We subtract and add the benchmark return, r_{bmk} , to obtain a return in excess of the benchmark, which is the active return, and the benchmark return:

$$r = \underbrace{(r - r_{bmk})}_{\text{Active Return}} + \underbrace{r_{bmk}}_{\text{Benchmark Return}}. \quad (14.1)$$

The poor active returns, $r_{active} = r - r_{bmk}$, were what the Norwegians were up in arms about in 2008 and the beginning of 2009. The benchmark returns, r_{bmk} , are the result of the asset allocation decision (later it will be a factor decision) taken by the Ministry of Finance. Benchmark returns are often called passive returns, especially if they can be implemented by passive index funds as in Norway’s case. The active returns, r_{active} , result from the decisions of NBIM to deviate from the benchmark.

Gary Brinson, Randolph Hood, and Gilbert Beebower, all practitioners, wrote a major study in 1986 showing that approximately 90% of a typical fund’s return

⁴ See chapters 6 to 9 for economic theory underlying these risk premiums.

⁵ Chapters 17 and 18 argue that hedge funds and private equity are not asset classes, respectively. Chapter 10 covers a factor decomposition of real estate.